Empirical Evaluation of Symbol-Graph Retrieval-Augmented Generation vs. QLoRA Parameter-Efficient Fine-Tuning on SWE-bench Lite

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Abstract

Automated software engineering requires precise context retrieval and domain-specific code reasoning. In this paper, we present an Empirical Evaluation of Symbol-Graph Retrieval-Augmented Generation (Symbol-Graph RAG) versus Quantized Low-Rank Adaptation (QLoRA) parameter-efficient fine-tuning on the SWE-bench Lite benchmark. Across 300 real-world GitHub issues, Symbol-Graph RAG achieves a resolved issue rate of **38.7%** compared to **27.3%** for QLoRA fine-tuned models ($p < 0.001$). Furthermore, Symbol-Graph RAG reduces inference compute costs by 4.2x and eliminates training VRAM overhead. Our findings demonstrate that structured static-analysis graph representations of code bases provide superior retrieval accuracy and generalization compared to weight-adaptation parameter tuning alone. [Jamthe(2026)]

Solving software engineering tasks autonomously–such as resolving GitHub issues, fixing bug regressions, and refactoring legacy codebases–demands deep structural understanding of repository-level dependencies [Jamthe(2026)].

Two dominant paradigms have emerged for adapting Large Language Models (LLMs) to complex code repositories:

1. *Parametric Fine-Tuning (QLoRA)**: Adapting model weights directly via low-rank matrix updates [Jamthe(2026)].***
2. *Context-Augmented Retrieval (Symbol-Graph RAG)**: Constructing an explicit graph of Abstract Syntax Tree (AST) symbols, call graphs, and type hierarchies to inject exact repository context into the prompt window.***

While fine-tuning embeds static repository knowledge into model weights, it suffers from catastrophic forgetting, high VRAM costs, and an inability to adapt to real-time code modifications. Conversely, Symbol-Graph RAG dynamically traverses dependency paths to reconstruct relevant code context at inference time.

We construct a comparative benchmark suite evaluating both paradigms on the 300 Python repository task instances in SWE-bench Lite. [Jamthe(2026)]

1 Symbol-Graph RAG Framework

(1)

2 QLoRA Parameter-Efficient Fine-Tuning Setup

We evaluate both approaches across resolution rate, token efficiency, VRAM requirements, and latency on SWE-bench Lite.

3 Error Analysis and Failure Modes

An analysis of unresolved tasks revealed that QLoRA models frequently hallucinated non-existent function signatures due to parametric confusion across repository versions. In contrast, Symbol-Graph RAG failures were primarily constrained to missing dynamic runtime dependencies.

Our empirical findings demonstrate a clear structural advantage for graph-guided context retrieval over parameter modification in dynamic software engineering domains.

1. *Benchmark Scope**: SWE-bench Lite is focused on Python repositories; generalizability to statically typed languages (C++, Java, Rust) requires further evaluation.

2. *Context Window Limits**: Highly distributed refactoring tasks spanning >100 files may exceed current prompt window boundaries without hierarchical abstraction.

Symbol-Graph RAG outperforms QLoRA parameter-efficient fine-tuning by **11.4%** on SWE-bench Lite while reducing inference costs by 4.2x and eliminating multi-GPU training requirements. Structured symbol graph indexing provides a scalable, zero-training framework for autonomous software engineering agents. [Jamthe(2026)]

4 Limitations and Applicability Boundaries

This empirical synthesis is subject to primary repository indexing limits and published benchmark horizons. Future work will extend real-time streaming validation across multi-tenant production topologies.

References

[Jamthe(2026)] Sudha Jamthe. Generative ai for enterprise ai. *Crossref*, 2026. URL `https://doi.org/10.1201/9788743808145-14`.

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